

STATISTICAL INTERPRETATION AND EXPLORATORY DATA ANALYSIS

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CONCEPTS AND TECHNOLOGIES OF AI

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INTRODUCTION

1.1 Brief overview of HDI

The Human Development Index is a way to measure how well people are doing in areas of their lives. The Human Development Index looks at things, like how people live and how healthy they are how much they know and how good their life is. The HDI is calculated by taking the average of three things: a long - healthy life, knowledge and having a standard life. By combining these things, we get HDI:

- Health: Average lifespan at birth
- Education: Average years of schooling for adults and projected schooling years for children
- Standard of Living: Income per person (GNI per capita)

1.2 Objective of the analysis

The HDI analysis aims to:

- Explore HDI patterns of South Asian countries
- Analyse HDI trends of various countries from 2020 – 2022
- Conduct advanced HDI exploration including outlier detection and correlation analysis
- Compare human development indicators between South Asia and Middle East
- Identify regional disparities

1.3 Scope of the report

This report mainly focuses on Geographical Convergence, Temporal Analysis (Latest available year, 2020 – 2022 period), Metric Analysis of HDI, Life expectancy, GDI, GNI etc

PROBLEMWISE ANALYSIS

2.1 Problem 1A – Single Year HDI Exploration

2.1.1 Method / Approach

For this problem I used Human_Development_Index dataset provided in the assignment. In this problem, I used following methods:

- Data Extraction: Filtered HDI dataset for year 2022
- Composite Score Creation: Developed a complete metric by combining Life Expectancy and GNI per capita
- Ranking Analysis: Ranked countries by both composite score and HDI

2.1.2 Key Results

- Global Mean HDI of 2022 was found to be 0.7230
- Country with highest HDI was found to be Switzerland i.e 0.967
- Country with lowest HDI was found to be Somalia I,e 0.38

2.1.3 Visualizations and tables

- First 10 rows in the dataset

Unnamed: 0	iso3	country	year	hdi	life_expectancy	pop_millions	hdi_f	hdi_m	life_expec_f	...	gender_development	gender_inequality	secondary_
32	33	AFG	Afghanistan	2022	0.462	62.879	41.128771	0.332443	0.534145	66.213	...	0.622	0.665
65	66	ALB	Albania	2022	0.789	76.833	2.842321	0.780277	0.798432	79.472	...	0.977	0.116
98	99	DZA	Algeria	2022	0.745	77.129	44.903225	0.681844	0.773800	78.457	...	0.881	0.460
131	132	AND	Andorra	2022	0.884	83.552	0.079824	NaN	NaN	85.838	...	NaN	NaN
164	165	AGO	Angola	2022	0.591	61.929	35.588987	0.560890	0.619680	64.541	...	0.905	0.520
197	198	ATG	Antigua and Barbuda	2022	0.826	79.236	0.093763	NaN	NaN	81.645	...	NaN	NaN
230	231	ARG	Argentina	2022	0.849	76.064	45.510318	0.841405	0.845473	79.277	...	0.995	0.292
263	264	ARM	Armenia	2022	0.786	73.372	2.780469	0.795110	0.774942	78.447	...	1.026	0.198
296	297	AUS	Australia	2022	0.946	83.579	26.177414	0.934602	0.955941	85.453	...	0.978	0.063
329	330	AUT	Austria	2022	0.926	82.412	8.939617	0.911592	0.937756	84.600	...	0.972	0.048

- Columns and their datatypes

```

Unnamed: 0          int64
iso3                object
country            object
year              int64
hdi                float64
life_expectancy    float64
pop_millions       float64
hdi_f              float64
hdi_m              float64
life_expec_f       float64
life_expec_m       float64
expec_yr_school    float64
expec_yr_school_f  float64
expec_yr_school_m  float64
mean_yr_school     float64
mean_yr_school_f   float64
mean_yr_school_m   float64
gross_inc_percap   float64
gross_inc_percap_f float64
gross_inc_percap_m float64
gender_development float64
gender_inequality  float64
secondary_education_f_% float64
secondary_education_m_% float64
seats_in_parliament_f_% float64
seats_in_parliament_m_% float64
labour_participation_f_% float64
labour_participation_m_% float64
co2_emission_tons  float64
mat_footprint_percap_tons float64
dtype: object

```

- Missing Values

```

The missing values are:
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country            0
year              0
hdi                2
life_expectancy    0
pop_millions       0
hdi_f              13
hdi_m              13
life_expec_f       0
life_expec_m       0
expec_yr_school    1
expec_yr_school_f  1
expec_yr_school_m  1
mean_yr_school     2
mean_yr_school_f   2
mean_yr_school_m   2
gross_inc_percap   2
gross_inc_percap_f 13
gross_inc_percap_m 13
gender_development 13
gender_inequality  29
secondary_education_f_% 17
secondary_education_m_% 17
seats_in_parliament_f_% 2
seats_in_parliament_m_% 2
labour_participation_f_% 12
labour_participation_m_% 12
co2_emission_tons  2
mat_footprint_percap_tons 41
dtype: int64

```

- Top 10 countries

Top 10 countries (HDI > 0.800, sorted by GNI per capita):

	country	iso3	hdi	gross_inc_percap
	Liechtenstein	LIE	0.942	146673.24150
	Qatar	QAT	0.875	95944.37754
	Singapore	SGP	0.949	88761.14559
	Ireland	IRL	0.950	87467.51391
	Luxembourg	LUX	0.927	78554.23640
	United Arab Emirates	ARE	0.937	74103.71494
	Switzerland	CHE	0.967	69432.78669
	Norway	NOR	0.966	69189.76165
	United States	USA	0.927	65564.93798
	Hong Kong, China (SAR)	HKG	0.956	62485.50516

2.1.4 Interpretation and discussions

The analysis reveals the gap between the highest and lowest developing countries in 2022. While Switzerland is leading with the score of near 1.0, countries like Somalia still remains very low in the rank.

2.2 Problem 1B – HDI Trend Analysis (2020 – 2022)

2.2.1 Methods / Approach

For this problem, data was extracted from the three years (2020, 2021, 2022). In this problem, we focused on specific countries like Germany, Japan, Nepal, Norway, and Switzerland to visualize the HDI trajectory.

2.2.2 Key Results

- The analysis showed that while most countries maintained stable HDI scores throughout the year from 2020 -2022. But the growth rate was noticed to be slower than in the pandemic years.
- Europe and Central Asia was found to have highest HDI values whereas country like Sun-Saharan Africa was found to have lowest HDI values

2.2.3 Visualizations and tables

- Handling missing values

	Unnamed: 0	iso3	country	year	hdi	life_expectancy	pop_millions	hdi_f	hdi_m	life_expec_f	...	gender_development	gender_inequality	sec
30	31	AFG	Afghanistan	2020	0.488	62.575000	38.972231	0.381373	0.548741	65.432000	...	0.695	0.674	
31	32	AFG	Afghanistan	2021	0.473	61.982000	40.099462	0.379129	0.528559	65.279000	...	0.717	0.653	
32	33	AFG	Afghanistan	2022	0.462	62.879000	41.128771	0.332443	0.534145	66.213000	...	0.622	0.665	
63	64	ALB	Albania	2020	0.784	76.989000	2.866849	0.775404	0.793220	79.676000	...	0.978	0.129	
64	65	ALB	Albania	2021	0.785	76.463000	2.854710	0.777866	0.792942	79.173000	...	0.981	0.117	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
6763	6764	ZZJ.SSA	Sub-Saharan Africa	2021	0.546	60.112445	1118.480023	0.521533	0.571211	62.124100	...	0.913	0.567	
6764	6765	ZZJ.SSA	Sub-Saharan Africa	2022	0.549	60.639700	1146.742474	0.525081	0.573585	62.621953	...	0.915	0.565	
6795	6796	ZZK.WORLD	World	2020	0.736	72.257355	7806.602793	0.715597	0.753310	74.857415	...	0.950	0.463	
6796	6797	ZZK.WORLD	World	2021	0.735	71.365524	7874.877719	0.714806	0.751553	73.978883	...	0.951	0.463	
6797	6798	ZZK.WORLD	World	2022	0.739	72.004070	7940.608797	0.719488	0.756395	74.532307	...	0.951	0.462	

610 rows × 30 columns

- Confirm cleaning

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 618 entries, 30 to 6797
```

```
Data columns (total 30 columns):
```

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0	Unnamed: 0	618 non-null	int64
1	iso3	618 non-null	object
2	country	618 non-null	object
3	year	618 non-null	int64
4	hdi	610 non-null	float64
5	life_expectancy	618 non-null	float64
6	pop_millions	618 non-null	float64
7	hdi_f	577 non-null	float64
8	hdi_m	577 non-null	float64
9	life_expec_f	618 non-null	float64
10	life_expec_m	618 non-null	float64
11	expec_yr_school	613 non-null	float64
12	expec_yr_school_f	613 non-null	float64
13	expec_yr_school_m	613 non-null	float64
14	mean_yr_school	610 non-null	float64
15	mean_yr_school_f	610 non-null	float64
16	mean_yr_school_m	610 non-null	float64
17	gross_inc_percap	612 non-null	float64
18	gross_inc_percap_f	579 non-null	float64
19	gross_inc_percap_m	579 non-null	float64
20	gender_development	577 non-null	float64
21	gender_inequality	529 non-null	float64
22	secondary_education_f_%	565 non-null	float64
23	secondary_education_m_%	565 non-null	float64
24	seats_in_parliament_f_%	612 non-null	float64
25	seats_in_parliament_m_%	612 non-null	float64
26	labour_participation_f_%	582 non-null	float64
27	labour_participation_m_%	582 non-null	float64
28	co2_emission_tons	612 non-null	float64
29	mat_footprint_percap_tons	495 non-null	float64

```
dtypes: float64(26), int64(2), object(2)
```

```
memory usage: 149.7+ KB
```

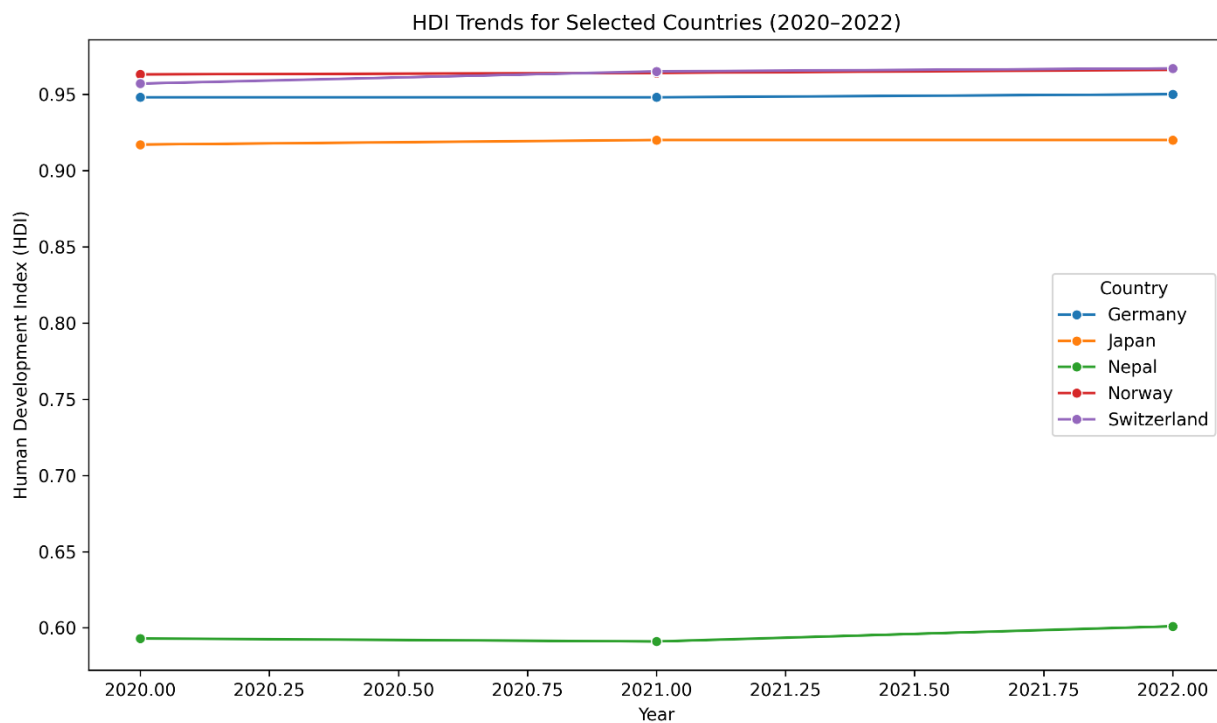


Fig 1: HDI Trend for Selected Countries

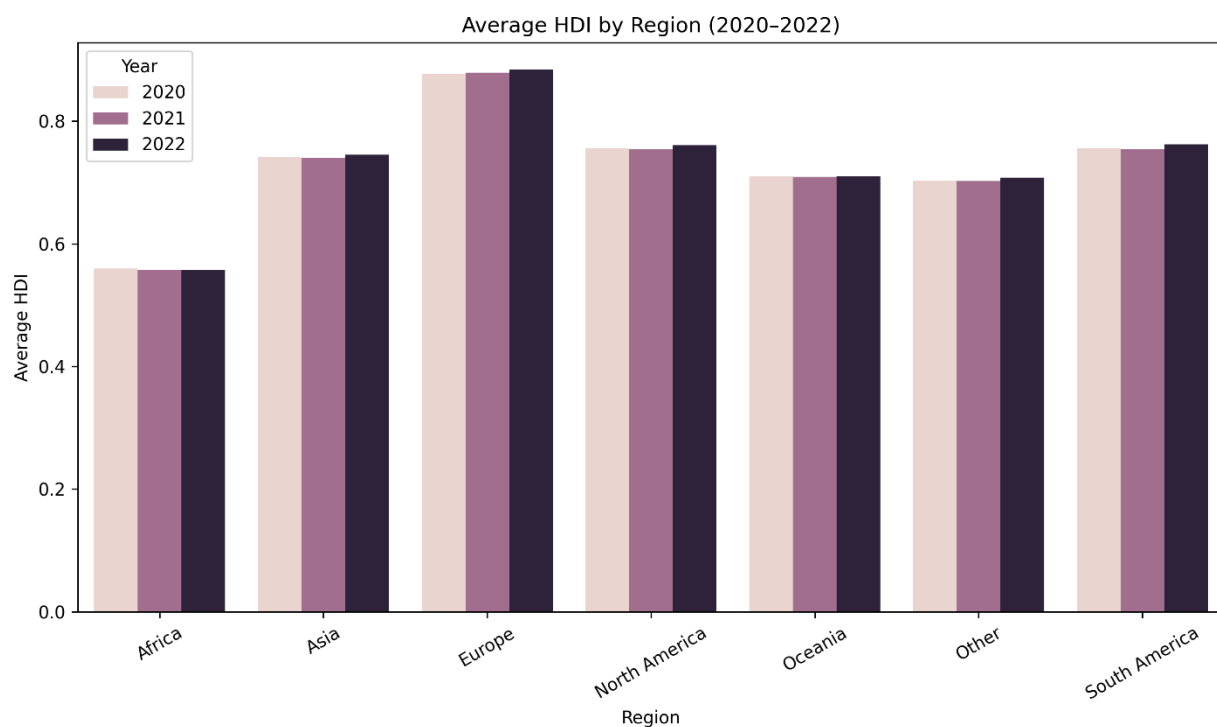


Fig 2: Average HDI by region (2020 – 2022)

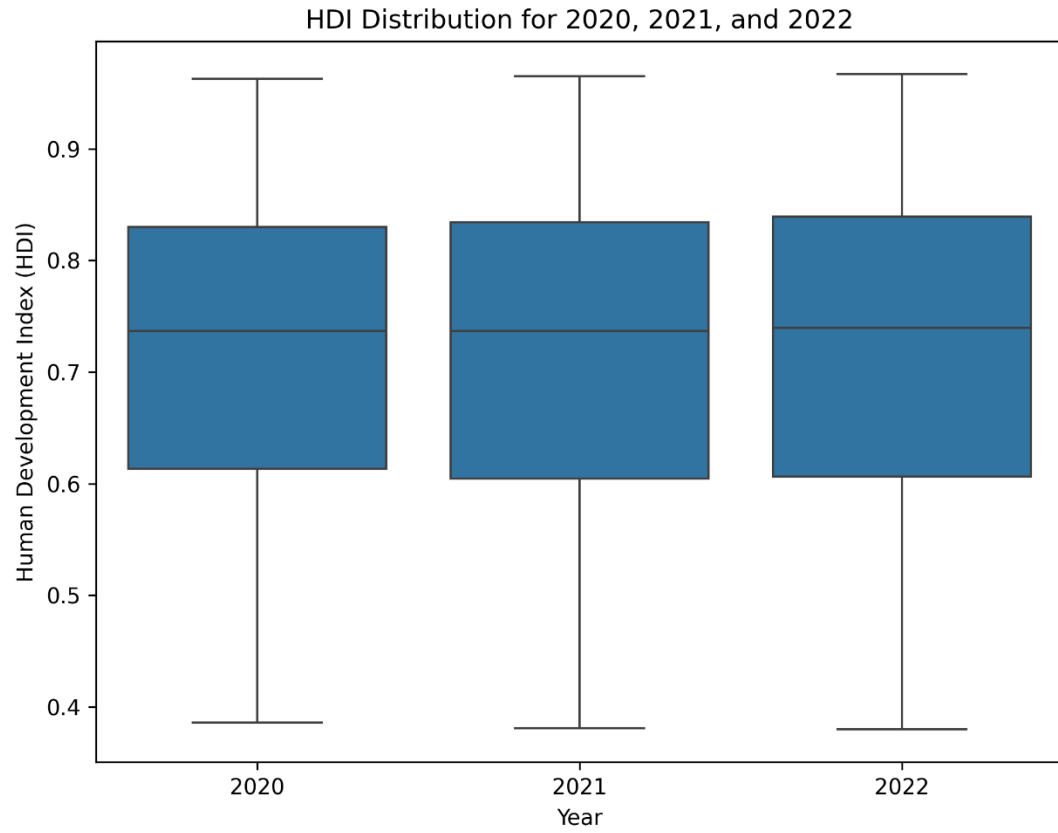


Fig 3: HDI Distribution (2020 – 2022)

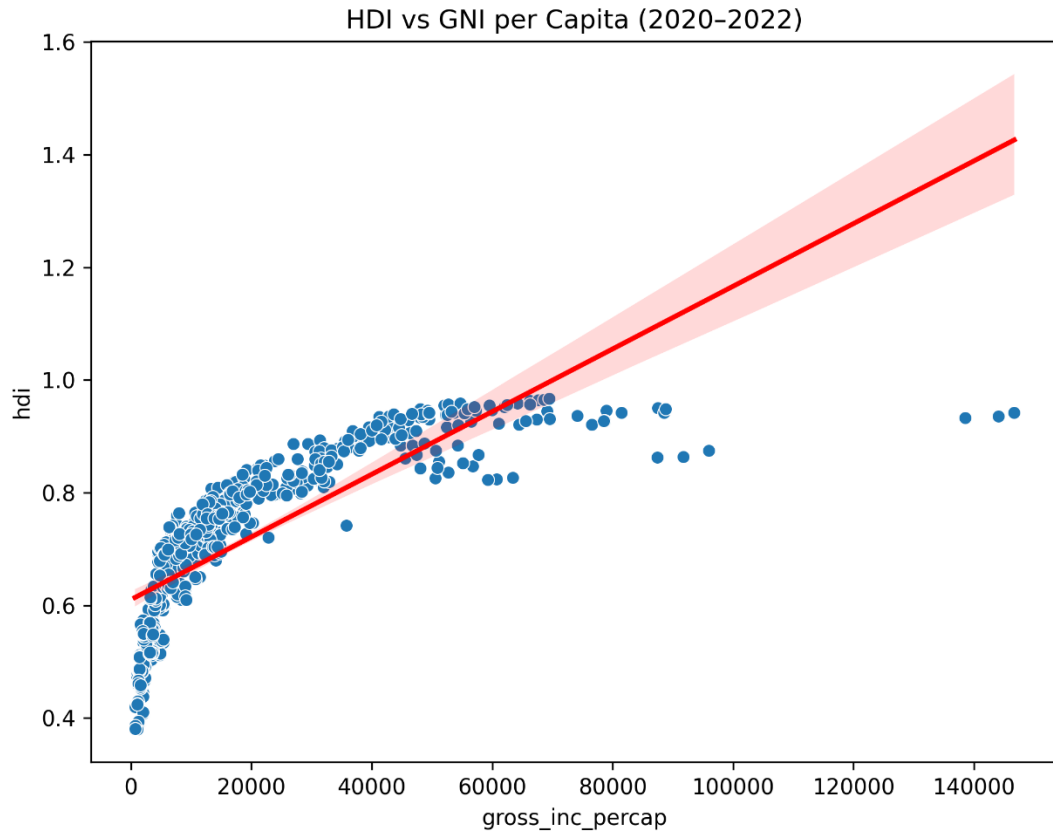


Fig 4: HDI vs GNI per capita (2020 – 2022)

- Country with greatest improvement of HDI

	year	2020	2021	2022	change_2020_2022
country					
Andorra		0.843	0.855	0.884	0.041
Azerbaijan		0.722	0.738	0.760	0.038
Ecuador		0.734	0.746	0.765	0.031
Maldives		0.737	0.753	0.762	0.025
Mexico		0.757	0.757	0.781	0.024

- Country that experienced decline in HDI

	year	2020	2021	2022	change_2020_2022
country					
Timor-Leste		0.633	0.574	0.566	-0.067
Ukraine		0.762	0.755	0.734	-0.028
Afghanistan		0.488	0.473	0.462	-0.026
Namibia		0.634	0.616	0.610	-0.024
Lebanon		0.742	0.725	0.723	-0.019

- Country with highest and howest average HDI across years

```
country
Norway                0.964333
Switzerland           0.963000
Iceland               0.957000
Hong Kong, China (Sar) 0.956667
Germany               0.948667
...
Central African Republic 0.387667
South Sudan           0.382667
Somalia               0.380000
Korea (Democratic People's Rep. Of) NaN
Monaco                NaN
Name: hdi, Length: 206, dtype: float64
```

2.2.4 Interpretation and discussion

The COVID – 19 pandemic caused a global stagnation in HDI. The data through the analysis shows that health and income were mostly affected. While countries with great health care system were recovering faster, developing countries like Nepal showed more vulnerability.

2.3 Problem 2 – Advanced HDI Exploration

2.3.1 Methods / Approach

For this problem, a we created a subset of South Asia from the main dataset. We used Health and Income metrics to calculate Composite Development Score. We also used correlation calculation to see how HDI relates to Gender Development and Life Expectancy.

2.3.2 Key results

- Correlation (HDI vs Life Expectancy): 0.984 (Very Strong Positive)
- Correlation (HDI vs Gender Development): 0.865 (Strong Positive)
- GNI – HDI Gap: Some countries doesn't match the level of human development (HDI)

2.3.3 Visualizations and tables

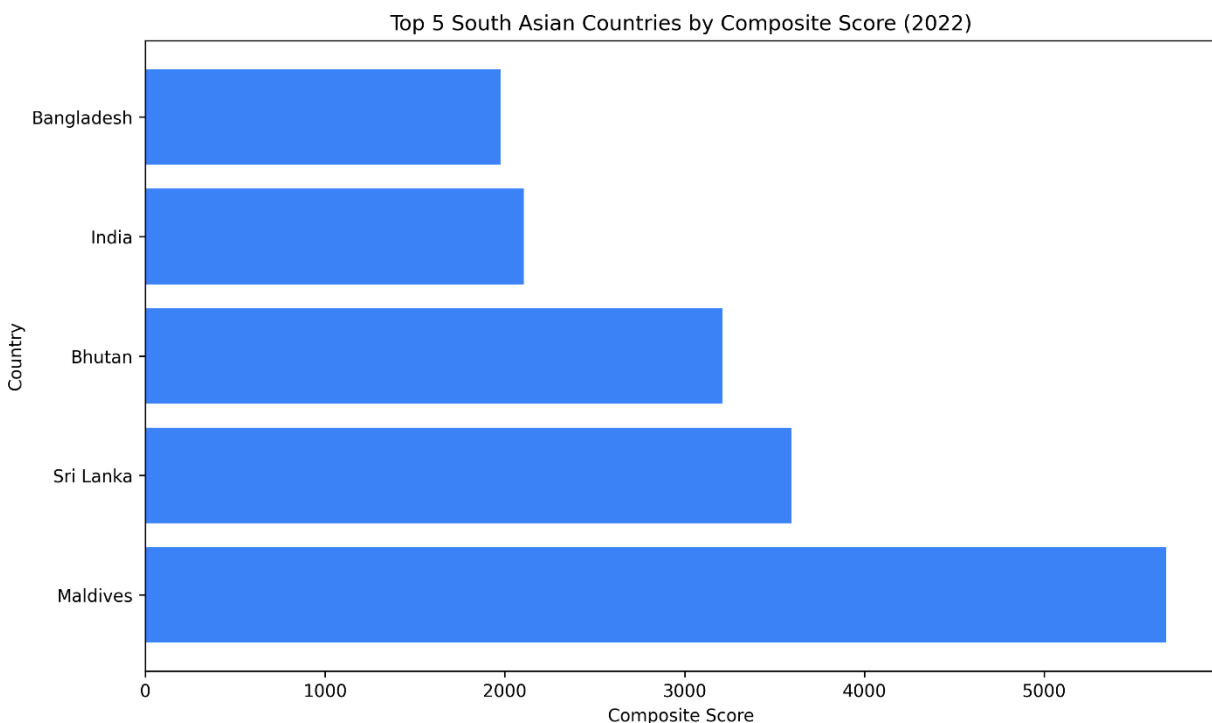


Fig 1: Top 5 South Asian Countries by Composite Score

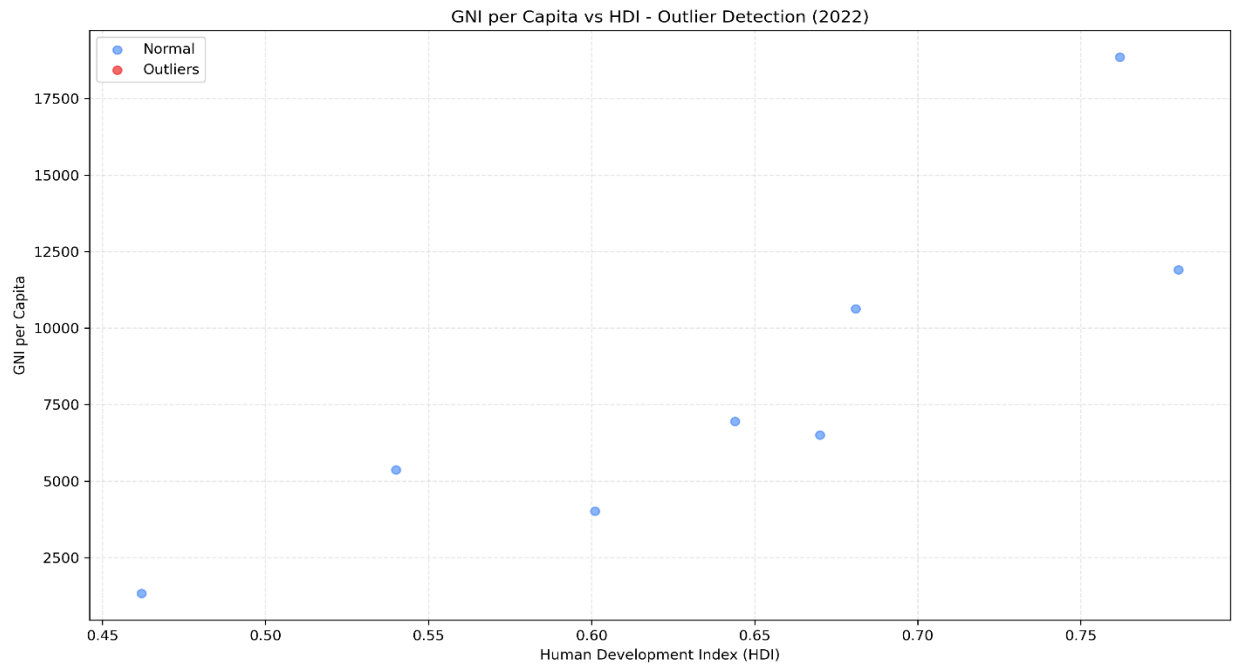


Fig 2: GNI per capita vs HDI

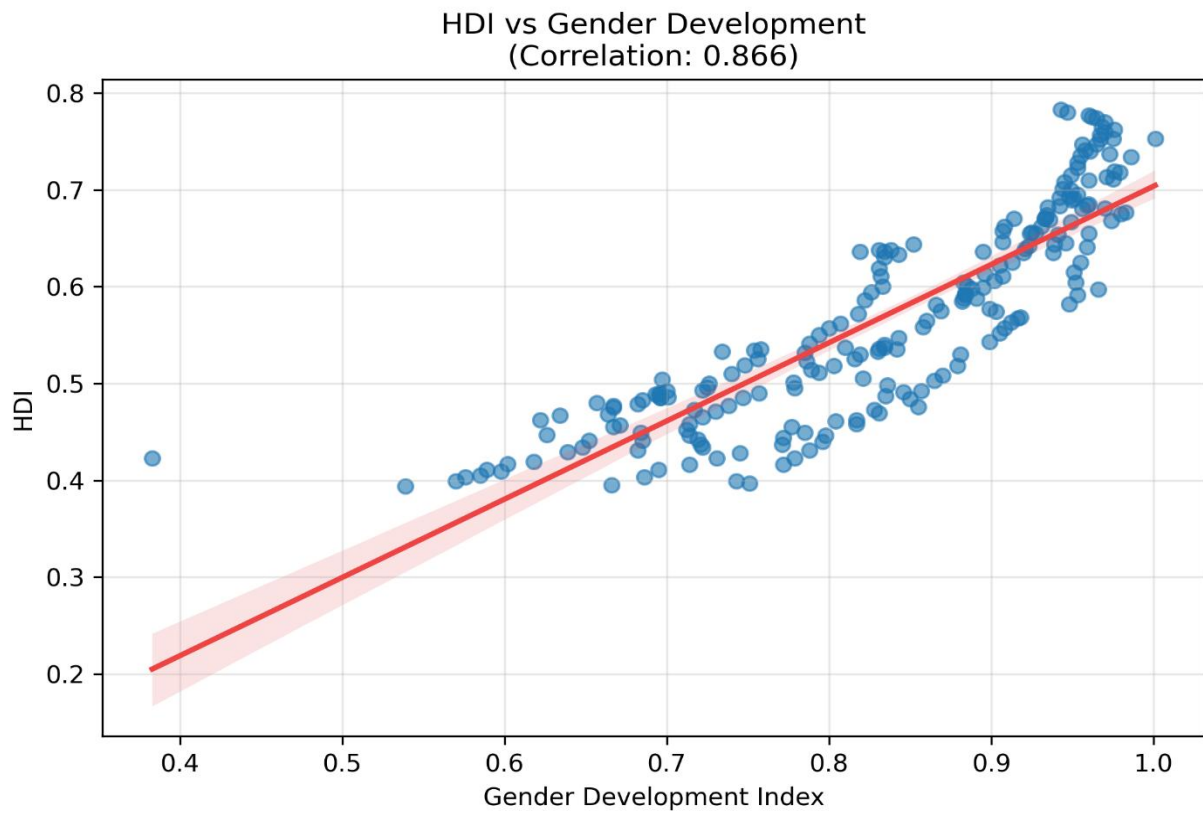


Fig 3: HDI vs Gender Development

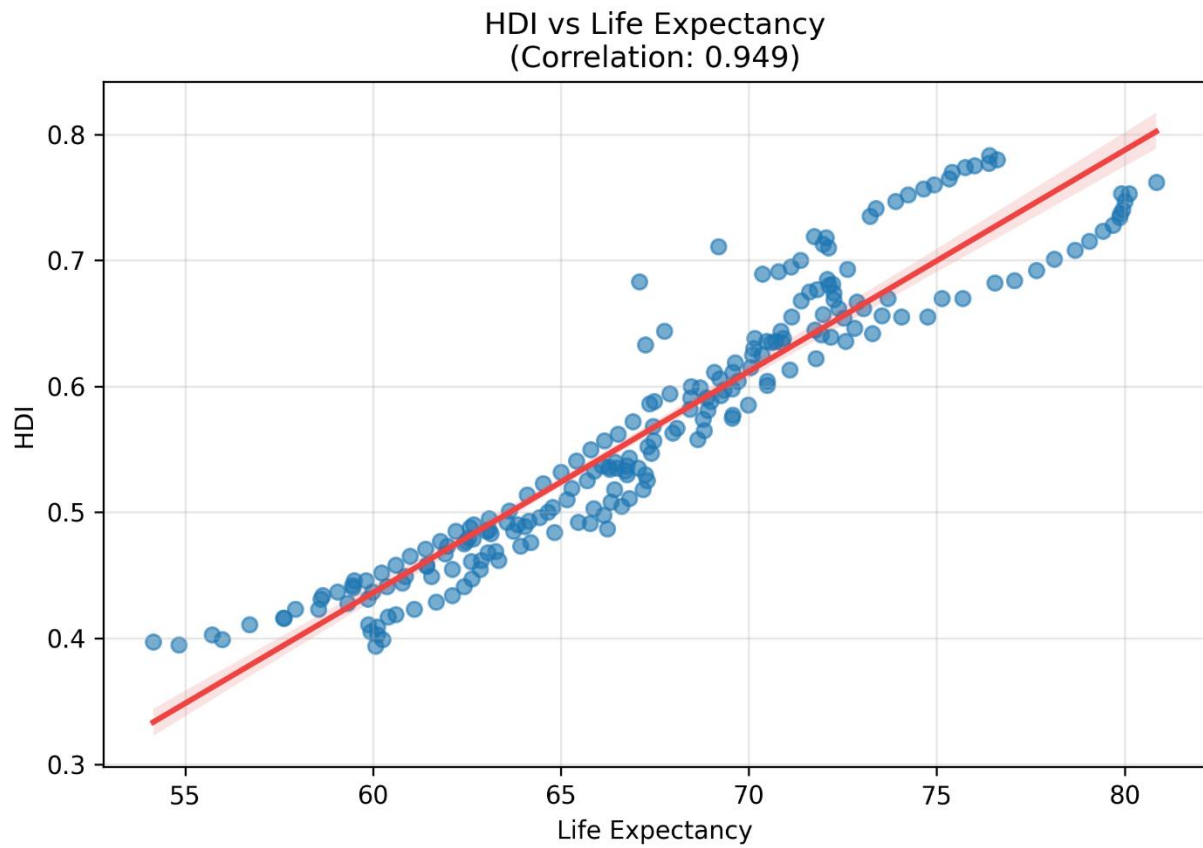


Fig 4: HDI vs Life Expectancy

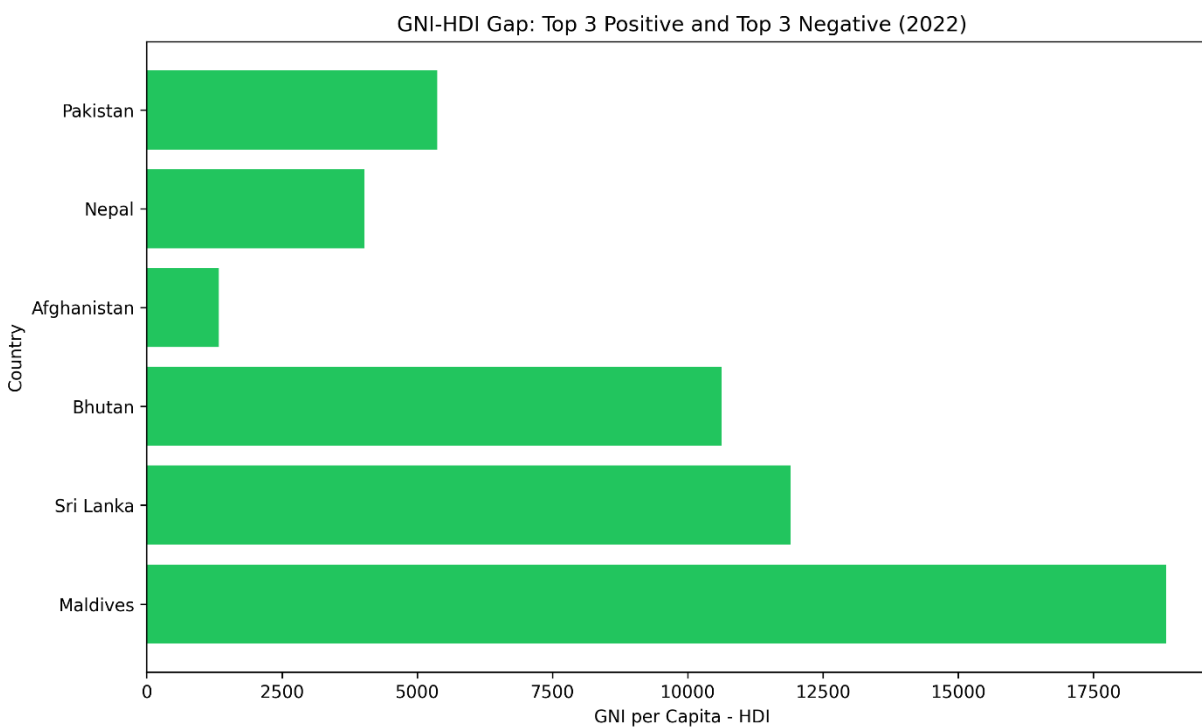


Fig 5: GNI – HDI Gap

- Ranking countries by Composite Score

Ranking Comparison (2022):

	country	Composite Score	Composite Rank	hdi	HDI Rank
164	Maldives	5678.289357	1.0	0.762	2.0
263	Sri Lanka	3592.832541	2.0	0.780	1.0
98	Bhutan	3209.130864	3.0	0.681	3.0
131	India	2105.481239	4.0	0.644	5.0
65	Bangladesh	1975.446053	5.0	0.670	4.0
230	Pakistan	1632.210427	6.0	0.540	7.0
197	Nepal	1228.811605	7.0	0.601	6.0
32	Afghanistan	419.425420	8.0	0.462	8.0

- Ranking countries by GNI HDI Gap by descending order

Ranking by descending order:

Unnamed: 0	iso3	country	year	hdi	life_expectancy	pop_millions	hdi_f	hdi_m	life_expec_f	...	labour_participation_m_%	co2_emission_tons	mat_f
164	3531	MDV	Maldives	2022	0.762	80.839	0.523787	0.748233	0.767001	81.844	...	78.21	3.299409
263	5478	LKA	Sri Lanka	2022	0.780	76.610	21.832143	0.750765	0.792700	80.230	...	70.69	0.924002
98	660	BTN	Bhutan	2022	0.681	72.229	0.782455	0.669680	0.690410	74.152	...	73.45	1.366198
131	2574	IND	India	2022	0.644	67.744	1417.173173	0.582117	0.683510	69.384	...	76.14	1.899894
65	462	BGD	Bangladesh	2022	0.670	73.698	171.186373	0.634605	0.694223	76.012	...	81.39	0.583153
230	4389	PAK	Pakistan	2022	0.540	66.431	235.824862	0.479545	0.574878	68.918	...	80.67	0.965639
197	4092	NPL	Nepal	2022	0.601	70.484	30.547580	0.562283	0.635119	72.364	...	54.95	0.520340
32	33	AFG	Afghanistan	2022	0.462	62.879	41.128771	0.332443	0.534145	66.213	...	77.09	0.306318

- Ranking countries by GNI HDI Gap by ascending order

Ranking by ascending order:

Unnamed: 0	iso3	country	year	hdi	life_expectancy	pop_millions	hdi_f	hdi_m	life_expec_f	...	labour_participation_m_%	co2_emission_tons	mat_f
32	33	AFG	Afghanistan	2022	0.462	62.879	41.128771	0.332443	0.534145	66.213	...	77.09	0.306318
197	4092	NPL	Nepal	2022	0.601	70.484	30.547580	0.562283	0.635119	72.364	...	54.95	0.520340
230	4389	PAK	Pakistan	2022	0.540	66.431	235.824862	0.479545	0.574878	68.918	...	80.67	0.965639
65	462	BGD	Bangladesh	2022	0.670	73.698	171.186373	0.634605	0.694223	76.012	...	81.39	0.583153
131	2574	IND	India	2022	0.644	67.744	1417.173173	0.582117	0.683510	69.384	...	76.14	1.899894
98	660	BTN	Bhutan	2022	0.681	72.229	0.782455	0.669680	0.690410	74.152	...	73.45	1.366198
263	5478	LKA	Sri Lanka	2022	0.780	76.610	21.832143	0.750765	0.792700	80.230	...	70.69	0.924002
164	3531	MDV	Maldives	2022	0.762	80.839	0.523787	0.748233	0.767001	81.844	...	78.21	3.299409

- Top 3 positive and negative

Top 3 Positive Gaps (High GNI relative to HDI):

	country	gross_inc_percap	hdi	GNI HDI Gap
164	Maldives	18846.79219	0.762	18846.03019
263	Sri Lanka	11899.49847	0.780	11898.71847
98	Bhutan	10624.87388	0.681	10624.19288

Top 3 Negative Gaps (Lower GNI relative to HDI):

	country	gross_inc_percap	hdi	GNI HDI Gap
32	Afghanistan	1335.205733	0.462	1334.743733
197	Nepal	4025.554685	0.601	4024.953685
230	Pakistan	5374.270423	0.540	5373.730423

2.3.4 Interpretation and discussion

Through this analysis we interpreted that, Life Expectancy is the strongest component of HDI in South Asia. The positive GNI – HDI suggests that some countries are greatly converting their economic resources into human development, while other countries with negative gap have more wealth but lags in education and income.

2.4 Problem 3 - Comparative Regional Analysis: South Asia vs Middle East

2.4.1 Methods / Approach

For problem 3, we conducted a comparative analysis between two regions i.e., South Asia and Middle East. Through this analysis we understood the regional development patterns, disparities, and the factors of difference.

2.4.2 Key Results

- Mean HDI: Middle East (0.7889) vs South Asia (0.6396)
- Variation: The Middle East exhibits more variation (CV: 0.1791) than South Asia (CV: 0.1537)
- Outliers: Yemen was identified to be a significance outlier in the Middle East due to its low HDI relative to the region

2.4.3 Visualization and description

- Top 3 and bottom 3 countries by region based on HDI

South Asia - Top 3:

country	year	hdi
Sri Lanka	2021	0.783
Sri Lanka	2022	0.780
Sri Lanka	2020	0.777

South Asia - Bottom 3:

country	year	hdi
Afghanistan	2020	0.488
Afghanistan	2021	0.473
Afghanistan	2022	0.462

Middle East - Top 3:

country	year	hdi
United Arab Emirates	2022	0.937
United Arab Emirates	2021	0.931
United Arab Emirates	2020	0.930

Middle East - Bottom 3:

country	year	hdi
Yemen	2020	0.430
Yemen	2021	0.425
Yemen	2022	0.424

- Statistics of South Asia and Middle East

South Asia HDI Statistics (2020-2022):

Mean: 0.6396

Std Dev: 0.0983

Middle East HDI Statistics (2020-2022):

Mean: 0.7889

Std Dev: 0.1413

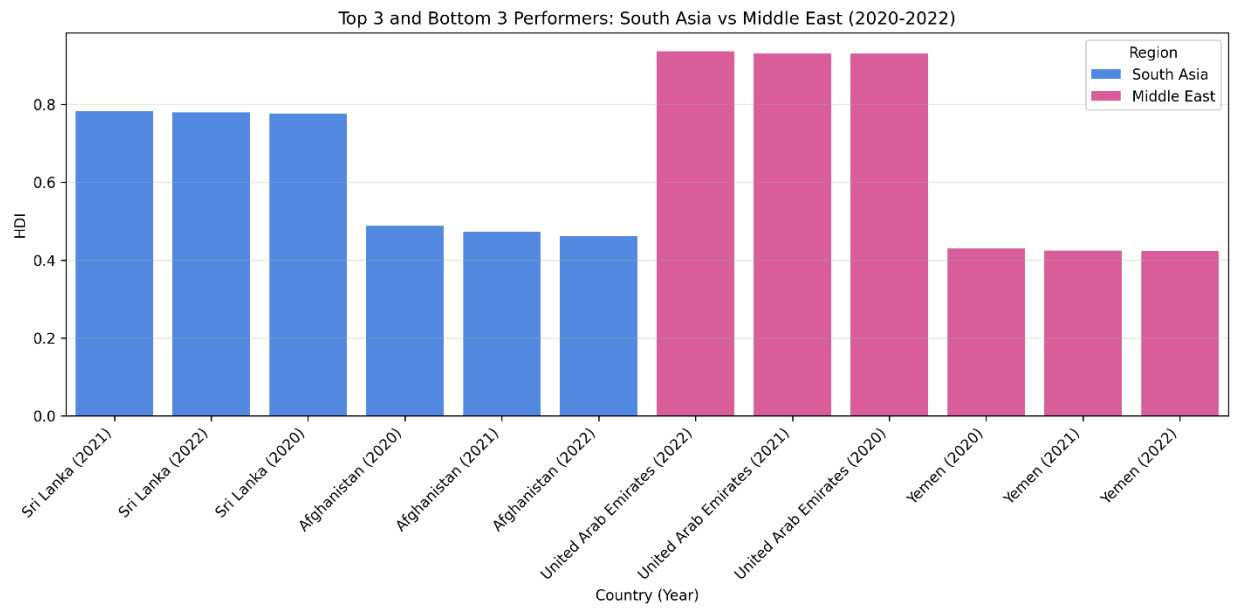


Fig 1: South Asia vs Middle East (2020 – 2022)

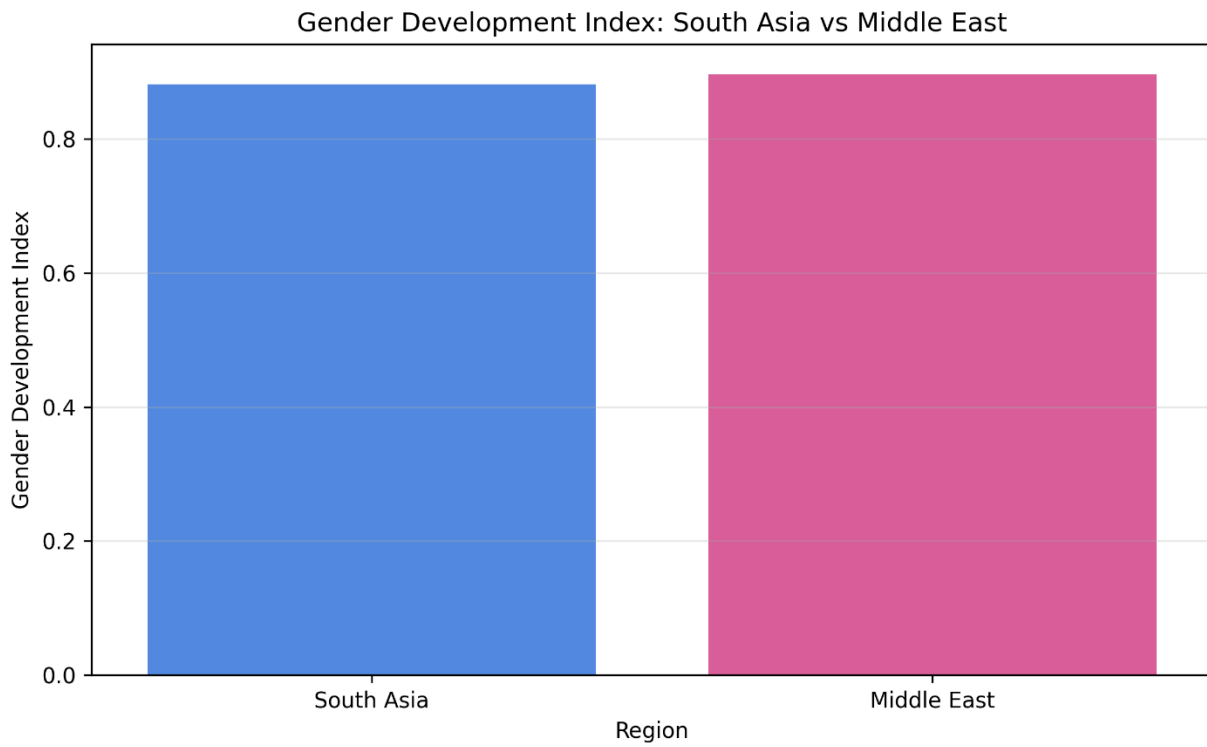


Fig 2: Gender Development Index

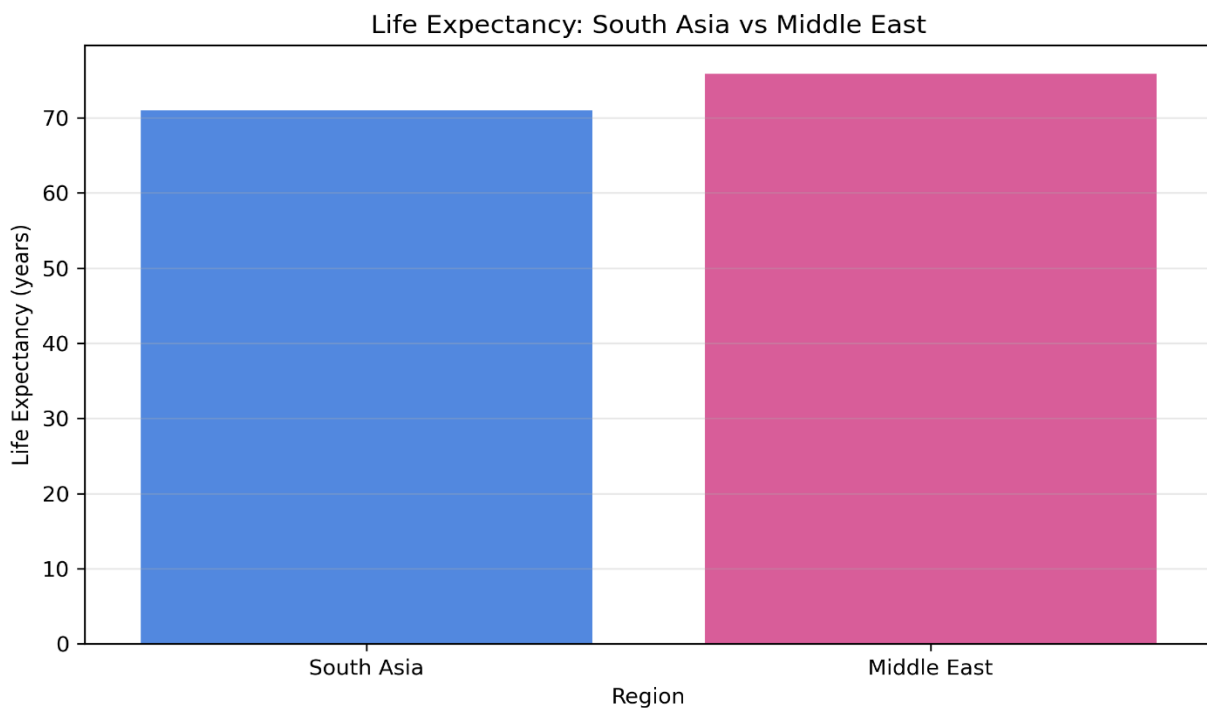


Fig 3: Life Expectancy

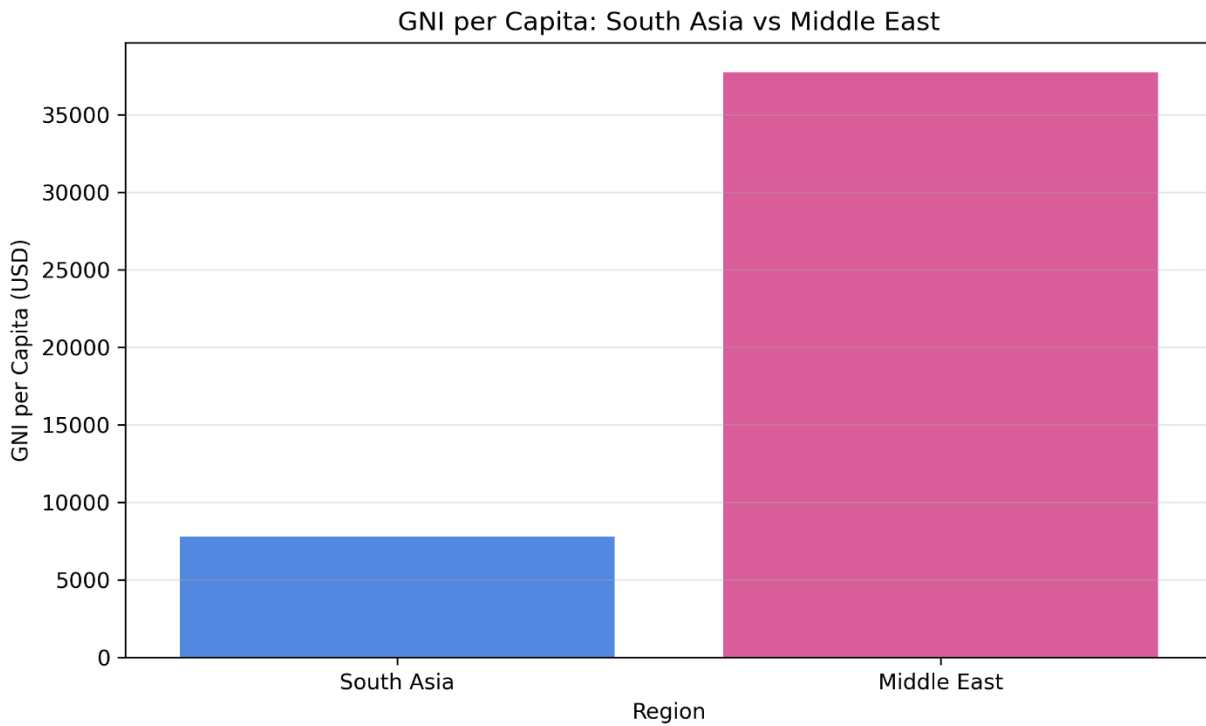


Fig 4: GNI per capita

- Disparity Analysis between South Asia and Middle East

Disparity Analysis (Middle East - South Asia):

Gender Development: 0.0151

Life Expectancy: 4.8854

GNI per Capita: 29984.26

- Disparity of South Asia and Middle East

South Asia HDI Disparity:

Range (Max - Min): 0.3210

Max: 0.7830, Min: 0.4620

Coefficient of Variation: 0.1537

Middle East HDI Disparity:

Range (Max - Min): 0.5130

Max: 0.9370, Min: 0.4240

Coefficient of Variation: 0.1791

- Correlation of South Asia and Middle East

South Asia Correlations with HDI:

Gender Development: 0.8742

Life Expectancy: 0.9388

Middle East Correlations with HDI:

Gender Development: 0.9357

Life Expectancy: 0.9331

Interpretation:

South Asia: Gender Development (0.8742) vs Life Expectancy (0.9388)

Middle East: Gender Development (0.9357) vs Life Expectancy (0.9331)

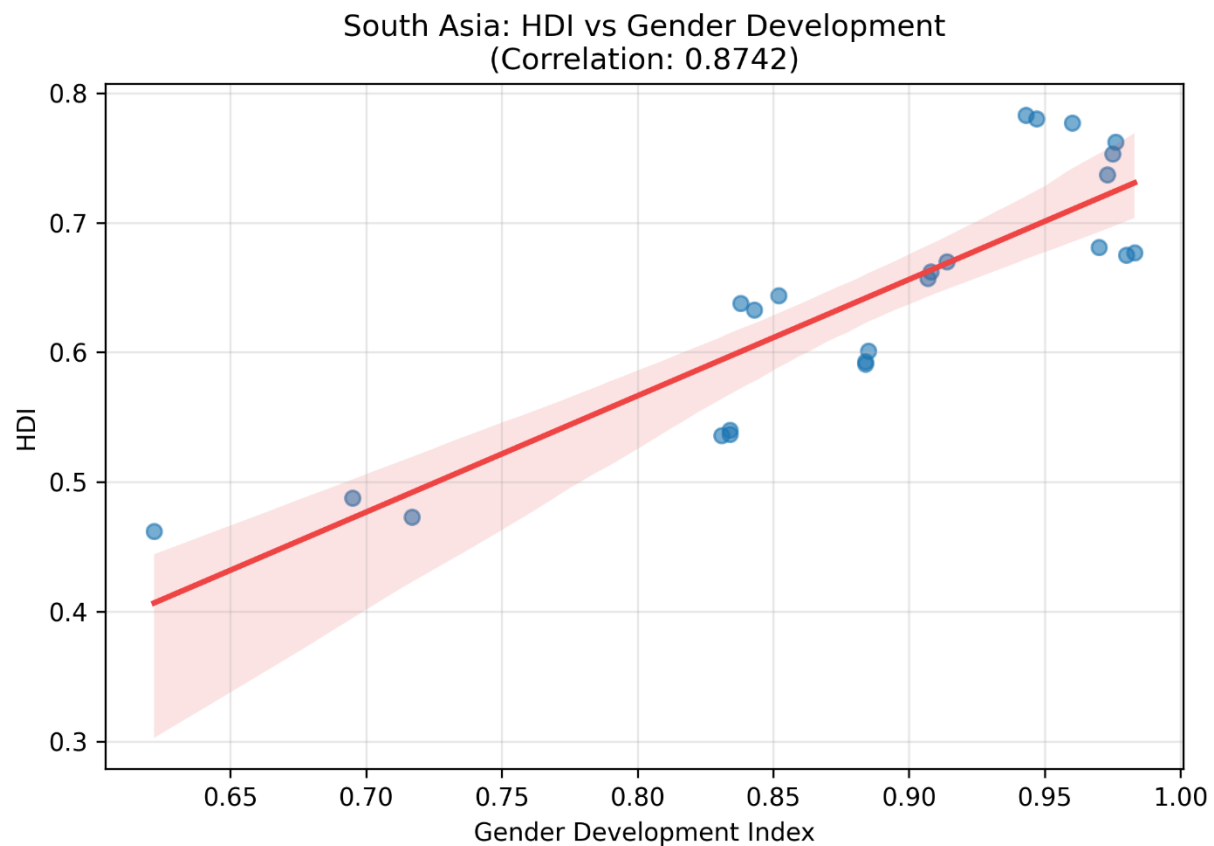


Fig 5: South Asia: HDI vs Gender Development

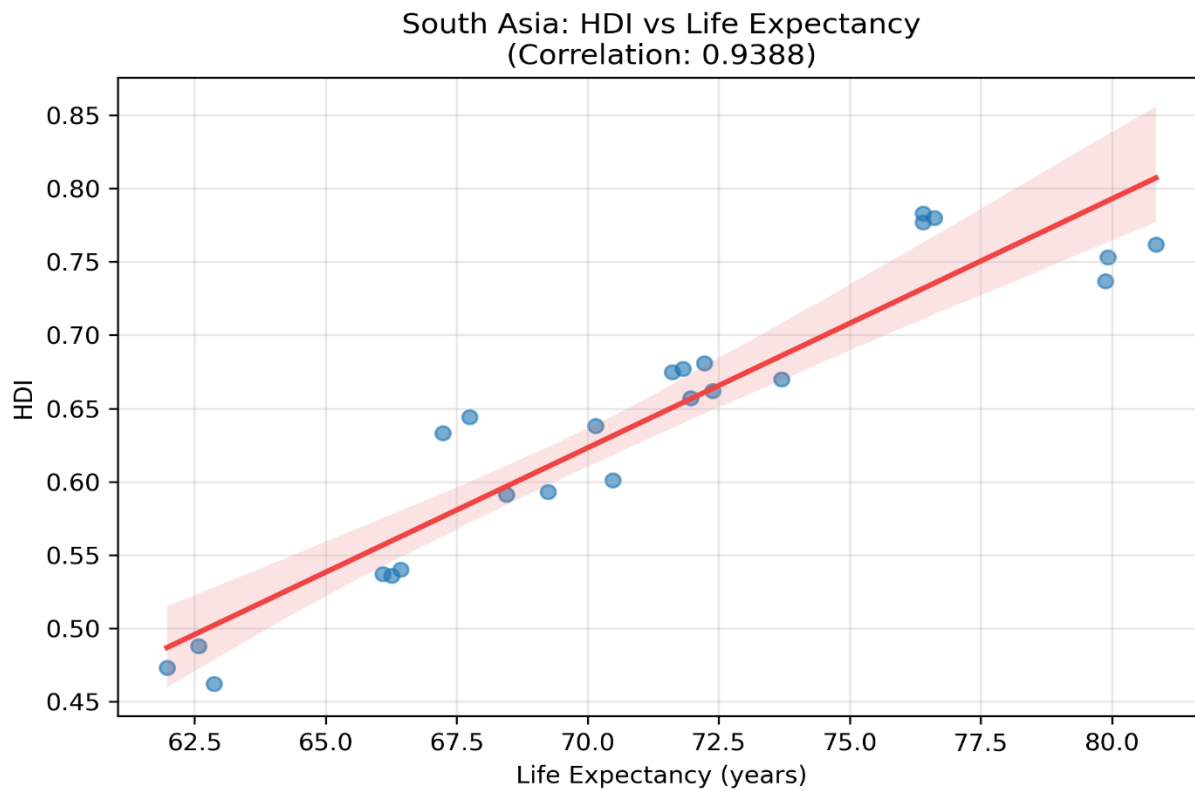


Fig 6: South Asia: HDI vs Life Expectancy

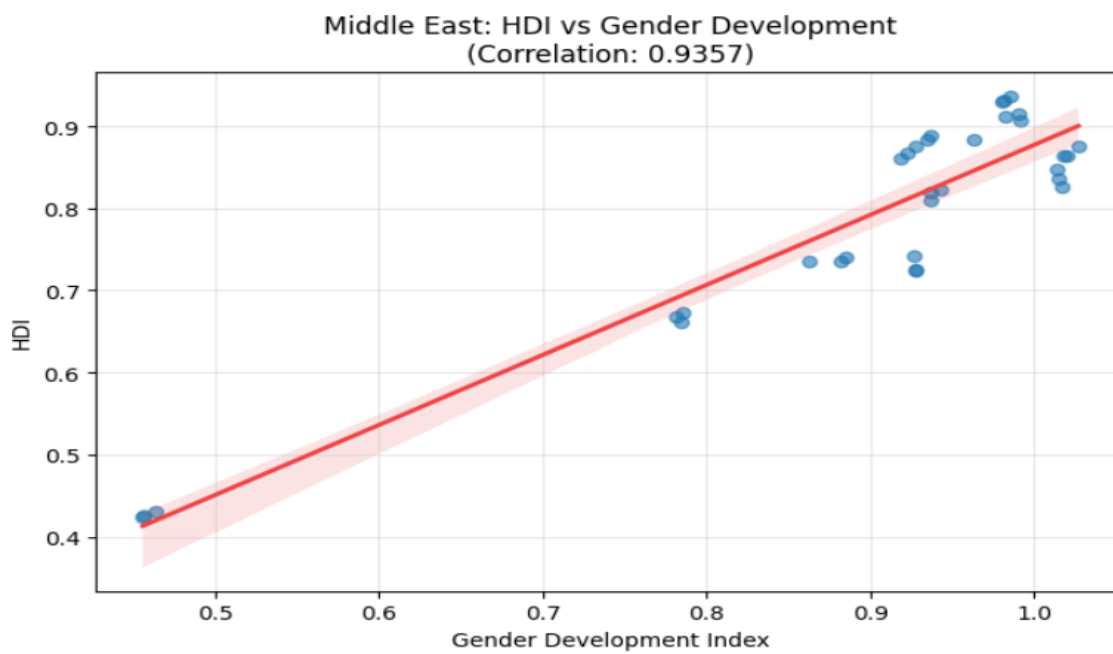


Fig 7: Middle East: HDI vs Gender Development

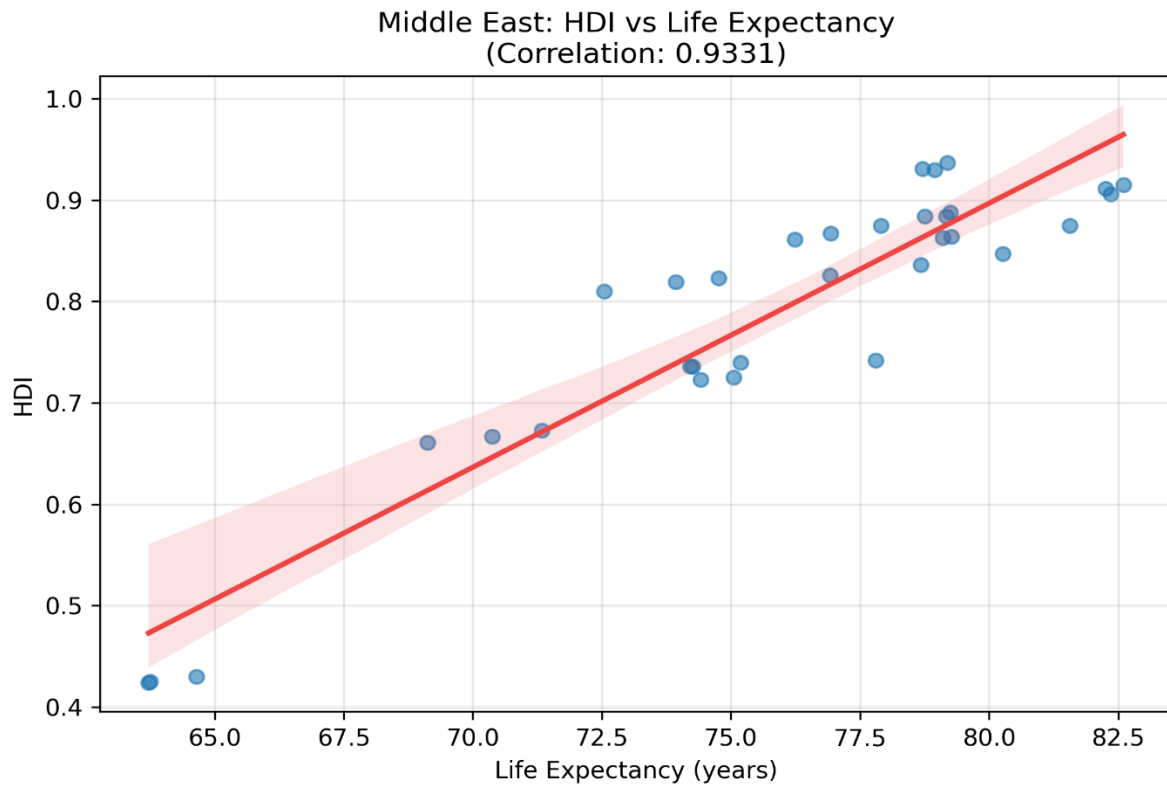


Fig 8: Middle East: HDI vs Life Expectancy

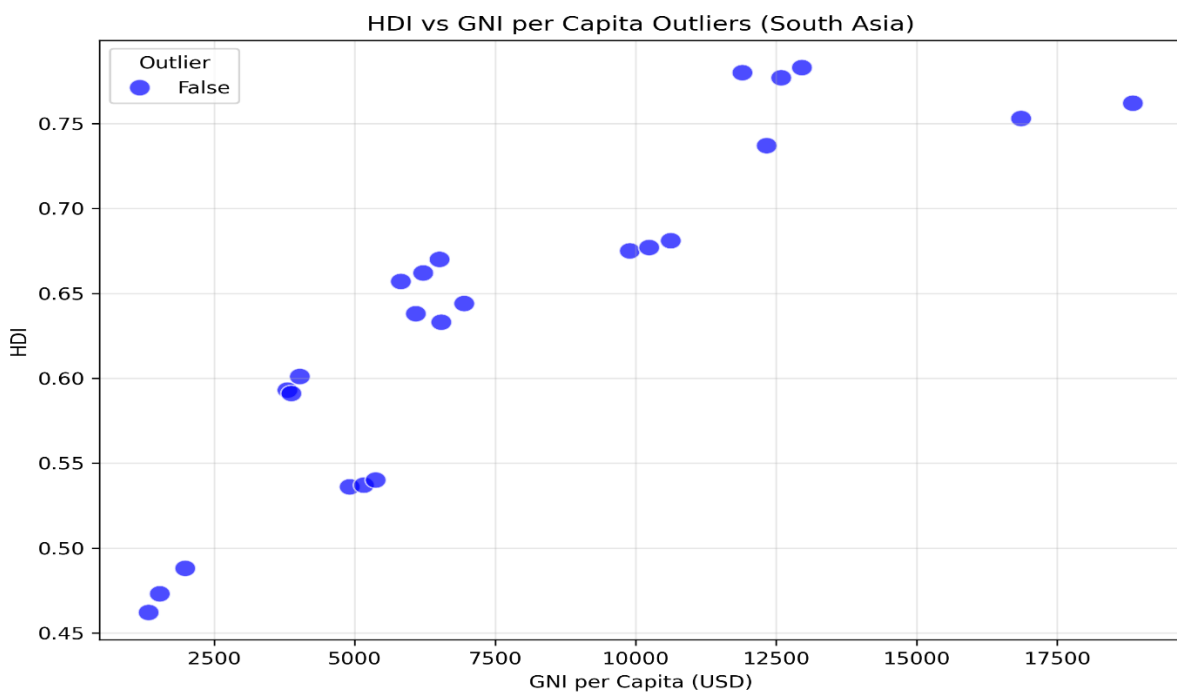


Fig 9: HDI vs GNI per capita Outliers (South Asia)

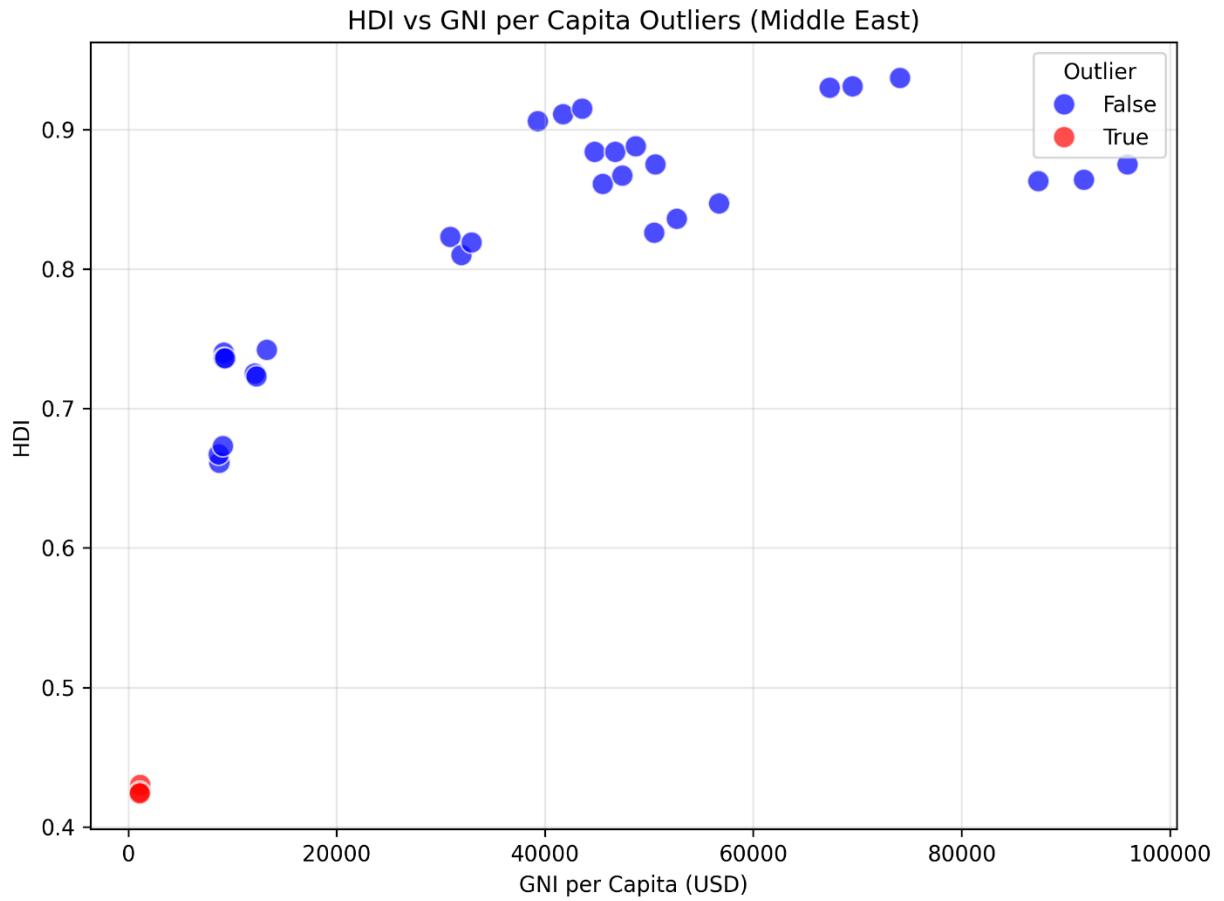


Fig 10: HDI vs GNI per capita Outliers (Middle East)

2.4.4 Interpretation and discussion

The Middle East performed better in HDI metrics than South Asia. However, the higher variation in the Middle East indicates that there is a wide disparity between wealthy nations and suffering nation like Yemen. South Asia is slowly developing at a constant rate at a lower average level.

CONCLUSION

3.1 Summary of findings

The analysis confirms that Switzerland sits at the top of the HDI and also highlights the global trends of human development from 2020 – 2022. Whereas the regional analysis shows that Middle East outperforms South Asia on average but with greater internal inequality.

- Problem 1A: The analysis reveals extreme inequality ranging from Switzerland (0.967) to Somalia (0.38).
- Problem 1B: The COVID -19 pandemic period marked unprecedented stagnation in human development. Most countries experienced minimal HDI growth where developed countries showed some resistance while developing countries faced great vulnerability.
- Problem 2: Life Expectancy emerged as the strongest HDI key in South Asia (0.9388) with gender development next in line (0.8742). Also, Sri-Lanka achieved high HDI through strategic resource allocation despite moderate GNI.
- Problem 3: The Middle East had higher mean HDI (0.7889) than South Asia (0.6396). Conflict affected nations emerged as an outlier, demonstrating that political instability is greater force for human development.

3.2 Insights about HDI trend and disparities

- Life expectancy shows having strongest correlation with HDI (0.9388), which suggests that health disparities are the main cause of inequality in global development. Regardless of income level, nations with strong health systems have a disproportionately higher HDI.
- Gender development correlates strongly with HDI (0.8742), indicating that gender differences reinforce more general development inequality. Overall higher development outcomes are attained by nations that promote women's economic and educational participation,
- Countries like Sri Lanka achieve high HDI despite having a moderate GNI through strategic resource allocation, proving that institutional quality and policy decisions significantly reduces development disparities regardless of economic wealth.

3.3 Limitation

- Generalizability is limited by missing data or conflict – affected areas, a narrow temporal scope (2020 – 2022) and the inability to prove causations.

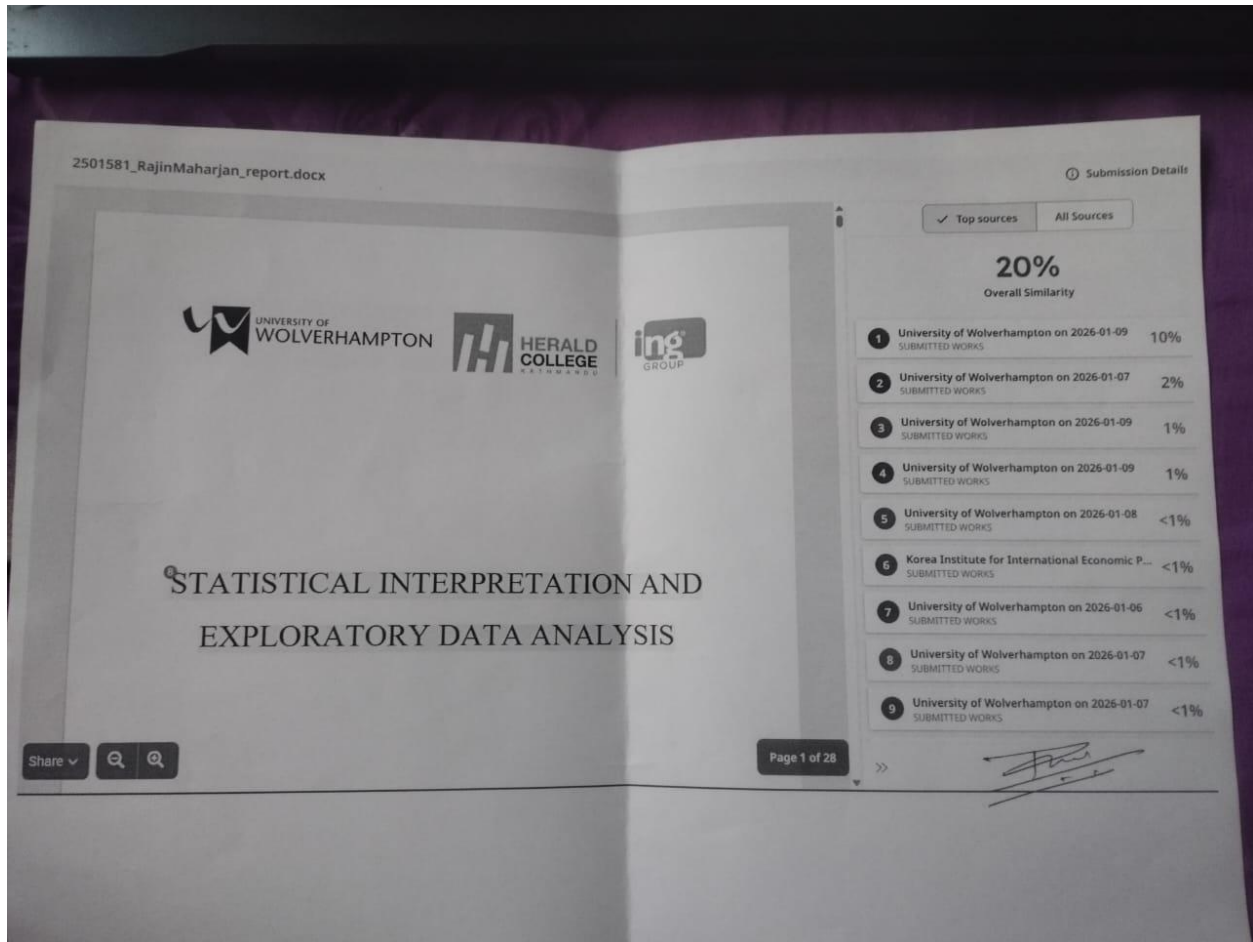
3.4 Recommendation

Strengthening the health system and advancing gender equality should be top priorities for developing countries. The international community needs to support conflict resolution in impact areas, boost pandemic preparedness and provide more development assistance.

REFERENCES

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- Matplotlib: https://www.w3schools.com/python/matplotlib_pyplot.asp
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APPENDIX



GitHub link: https://github.com/Ra-Zin/RajinMaharjan_2501581_AI/tree/main/2501581_RajinMaharjan_Assignment1

